

File



2012 Assessment Task 2

MATHEMATICS

Extension 1

Year 12

Time allowed - 60 minutes (plus 5 minutes reading time)

Topics: Logarithmic and Exponential Functions, Trigonometric Functions I and II

Instructions

Name and Class _____

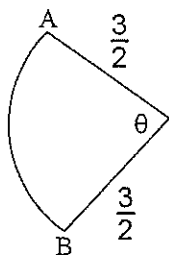
- Attempt all questions.
- Questions are of equal value.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Write on one side of the paper only.
- Questions do not necessarily appear in order of difficulty.
- Diagrams are not to scale
- A table of standard integrals is attached

Question One (12 marks)

a) Evaluate $\log_2 8$ [1]

b) Find the equation of the tangent to the curve $y = e^{2x}$ at the point where $x = 1$ [3]

c) The area of the sector below is $\frac{3\pi}{8} \text{ cm}^2$ and its radius is $\frac{3}{2} \text{ cm}$



i) Find the size of the angle θ in radians [2]

ii) Hence find the exact length of arc AB [1]

d) Find $\int \frac{x-1}{x^2-2x+3} dx$ [2]

e) For what value of x is $\log_3(x+1) - \log_3 x = 2$ [3]

Question Two (12 Marks)

a) Find the derivatives of:

i) $y = e^{2\sin x}$ [1]

ii) $y = \sec x$ [1]

b) If $\tan \frac{\theta}{2} = 2$, find the exact value of $\sin \theta$ [2]

c) i) Sketch $y = 3 \sin \frac{x}{2}$ for $-\pi \leq x \leq \pi$ [2]

ii) Hence find the area bounded by the curve, the X-axis and $x = \pm \frac{\pi}{2}$

d) Solve $2 \sin x \cos x - \cos x = 0$ for $0 \leq x \leq 2\pi$ [3]

Question Three (12 marks)

a) i) Find $\frac{d}{dx}(\log_e x)^2$ [1]

ii) Hence evaluate $\int_1^2 \frac{\log_e x}{x} dx$ [2]

b) Without the use of a calculator show that $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$ [3]

c) i) Sketch the curve $y = \log_e(x+2)$ showing all key features [2]

ii) The area bounded by $y = \log_e(x+2)$ and the axes is rotated about the Y- axis. Find the exact volume of the solid generated [4]

Question Four (12 Marks)

a) Find $\int e^{-3x} dx$ [1]

b) Find $\int \sin^2 x dx$ [2]

c) Find the sum of $\log_a \frac{1}{x} + \log_a \frac{1}{x^2} + \log_a \frac{1}{x^3} + \dots + \log_a \frac{1}{x^6}$ for $x > 1$ [2]

d) Evaluate $\lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{x}$ [2]

e) i) Write $2 \sin x + \cos x$ in the form $r \sin(x + \alpha)$ [2]

ii) Hence solve $2 \sin x + \cos x = 1$ for $0^\circ \leq x \leq 360^\circ$ [3]

Question Five (12 Marks)

- a) Find the size of the acute angle between the lines with
gradients $m_1 = 3$ and $m_2 = \frac{7}{2}$ [2]

- b) i) Show that $\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$ [2]

- ii) Hence find the exact value of $\cot 15^\circ$ [1]

- c) Find $\int 2x^2 e^{x^3} dx$ [2]

- d) Show that $\frac{d}{dx} \log_e \left(\frac{1 + \sin x}{\cos x} \right) = \sec x$ [2]

- e) Find $\int \sqrt{1 - \sin 2x} dx$ given that $0 < x < \frac{\pi}{4}$ (justify your answer) [3]

Question One (12 marks)

a) $\log_2 8 = x$ i.e. $2^x = 8$
 $x = 3$

b) $y = e^{2x}$
 $\frac{dy}{dx} = 2e^{2x}$

when $x=1$

$m = 2e^2$, $y = e^2$

$y - y_1 = m(x - x_1)$

$y - e^2 = 2e^2(x - 1)$

$y - e^2 = 2xe^2 - 2e^2$

$y = 2xe^2 - e^2$

c) i) $A = \frac{1}{2} r^2 \theta$

i.e. $\frac{3\pi}{8} = \frac{1}{2} \left(\frac{3}{2}\right)^2 \theta$

$\frac{3\pi}{8} = \frac{9\theta}{8}$

$\theta = \frac{3\pi}{8} \times \frac{8}{9}$

$= \frac{\pi}{3}$

d) $\int \frac{x-1}{x^2-2x+3} dx = \frac{1}{2} \int \frac{2x-1}{x^2-2x+3} dx$
 $= \frac{1}{2} \log_e (x^2-2x+3) + c$
 or $\log_e \sqrt{x^2-2x+3}$

e) $\log_3 (x+1) - \log_3 x = 2$

$\log_3 \frac{x+1}{x} = 2$

$3^2 = \frac{x+1}{x}$

$9 = \frac{x+1}{x}$

$9x = x+1$

$8x = 1$ $x = \frac{1}{8}$

Question Two (12 marks)

a) $y = e^{2\sin x}$

i) $\frac{dy}{dx} = (2\cos x)(e^{2\sin x})$

ii) $y = \sec x$
 $= (\cos x)^{-1}$

$\frac{dy}{dx} = -(\cos x)^{-2} (-\sin x)$

$= \frac{\sin x}{\cos^2 x}$

$= \tan x \sec x$

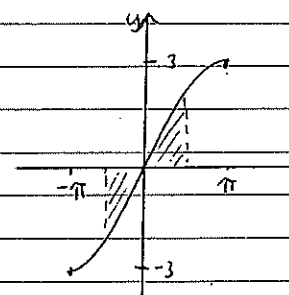
b) $\tan \frac{\theta}{2} = 2$ i.e. $x = 2$

$\sin \theta = \frac{2x}{1+x^2}$

$= \frac{4}{5}$

c) i) $y = 3\sin \frac{x}{2}$

amp = 3 period = $\frac{2\pi}{\frac{1}{2}}$
 $= 4\pi$



ii) $A = 2 \int_0^{\pi} 3\sin \frac{x}{2} dx$

$= [6x - 2\cos \frac{x}{2}]_0^{\pi}$

$= -12 [\cos \frac{\pi}{2}]_0^{\pi}$

$= -12 [\cos \frac{\pi}{2} - \cos 0]$

$= -12 [\frac{1}{\sqrt{2}} - 1]$

$= -6\sqrt{2} + 12 \text{ units}^2$

$(12 - 6\sqrt{2}) \text{ units}^2$

Q2 continued

$$\begin{aligned}
 \text{d)} \quad 2\sin x \cos x - \cos x &= 0 \\
 \cos x (2\sin x - 1) &= 0 \\
 \cos x = 0 &\quad \text{or} \quad \sin x = \frac{1}{2} \\
 \text{for } \sin x = \frac{1}{2} & \quad x = \frac{\pi}{6}, \frac{5\pi}{6} \\
 \text{for } \cos x = 0 & \quad x = \frac{\pi}{2}, \frac{3\pi}{2}
 \end{aligned}$$

Question 3 (12 marks)

$$\begin{aligned}
 \text{a) i)} \quad \frac{d}{dx} (\log x)^2 &= 2 \log x \times \frac{1}{x} \\
 &= \frac{2 \log x}{x}
 \end{aligned}$$

$$\text{ii)} \quad \int_1^2 \frac{\log x}{x} dx$$

$$\text{Now } \frac{d}{dx} (\log x)^2 = \frac{2 \log x}{x}$$

$$\therefore \int \frac{2 \log x}{x} = (\log x)^2$$

$$\therefore \int \frac{\log x}{x} = \frac{1}{2} (\log x)^2 + C$$

$$\int_1^2 \frac{\log x}{x} dx = \frac{1}{2} [(\log x)^2]_1^2$$

$$= \frac{1}{2} [(\log 2)^2 - (\log 1)^2]$$

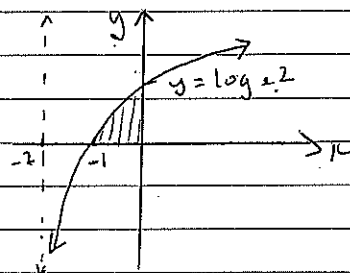
$$= \frac{1}{2} (\log 2)^2$$

$$\begin{aligned}
 \text{b)} \quad \text{let } x &= \tan^{-1} \frac{1}{2}, \quad y = \tan^{-1} \frac{1}{3} \\
 \text{ie } \tan x &= \frac{1}{2} \quad \text{ie } \tan y = \frac{1}{3} \\
 \text{prove } x+y &= \frac{\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 \tan(x+y) &= \frac{\tan x + \tan y}{1 - \tan x \tan y} \\
 &= \left(\frac{1}{2} + \frac{1}{3}\right) \div \left(1 - \frac{1}{2} \times \frac{1}{3}\right) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{ie } \tan(x+y) &= 1 \\
 x+y &= \tan^{-1} 1 \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$\text{c) i) } y = \log_2(x+2)$$



$$\text{ii) } y = \log_2(x+2) \Rightarrow x = 2^y - 2$$

$$\begin{aligned}
 V &= \pi \int_0^{\log_2 2} (2^y - 2)^2 dy \\
 &= \pi \int_0^{\log_2 2} (2^{2y} - 4 \cdot 2^y + 4) dy \\
 &= \pi \left[\frac{1}{2} 2^{2y} - 4 \cdot 2^y + 4y \right]_0^{\log_2 2} \\
 &= \pi \left[\frac{1}{2} 2^{2 \log_2 2} - 4 \cdot 2^{\log_2 2} + 4 \log_2 2 - \frac{1}{2} + 4 - 0 \right] \\
 &= \pi \left[\frac{1}{2} \times 4 - 4 \times 2 + 4 \log_2 2 + 3 \frac{1}{2} \right] \\
 V &= \pi \left[4 \log_2 2 - \frac{5}{2} \right] \text{ units}^3
 \end{aligned}$$

Question 4 (12 marks)

$$\text{a) } \int e^{-3x} dx = -\frac{1}{3} e^{-3x} + C$$

$$\begin{aligned}
 \text{b) } \int \sin^2 x dx &= \frac{1}{2} \int (1 + \cos 2x) dx \\
 &= \frac{1}{2} x - \frac{1}{4} \sin 2x + C
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \log_a \frac{1}{x} + \log_a \frac{1}{x^2} + \log_a \frac{1}{x^3} + \dots + \log_a \frac{1}{x^6} \\
 = \log_a x^{-1} + \log_a x^{-2} + \log_a x^{-3} + \dots + \log_a x^{-6} \\
 = -\log_a x - 2 \log_a x - 3 \log_a x + \dots - 6 \log_a x \\
 = -21 \log_a x
 \end{aligned}$$

$$d) \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= 1$$

$$e) i) 2 \sin x + \cos x = r \sin(x + \alpha)$$

$$r = \sqrt{1+1}$$

$$= \sqrt{2}$$

$$2 \sin x + \cos x = \sqrt{2} \sin(x + \alpha)$$

$$ii) \frac{2}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \sin x \cos \alpha + \cos x \sin \alpha$$

$$\cos \alpha = \frac{2}{\sqrt{2}}$$

$$\sin \alpha = \frac{1}{\sqrt{2}}$$

$$\tan \alpha = \frac{1}{2} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{1}{2}$$

$$\alpha = 26^\circ 34'$$

$$2 \sin x + \cos x = 1$$

$$\text{i.e. } \sqrt{2} \sin(x + 26^\circ 34') = 1$$

$$\sin(x + 26^\circ 34') = \frac{1}{\sqrt{2}}$$

$$(x + 26^\circ 34') = \frac{1}{\sqrt{2}} \quad (26^\circ 34' \leq x + 26^\circ 34' \leq 386^\circ 34') \quad (2)$$

$$x + 26^\circ 34' = 26^\circ 34', 153^\circ 26', 386^\circ 34'$$

$$x = 0, 126^\circ 52', 360^\circ \quad (3)$$

Question Five (12 marks)

$$a) \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{(3 - \frac{7}{2})}{(1 + \frac{21}{2})} \right|$$

$$= \frac{1}{23}$$

$$\theta = 2^\circ$$

$$b) i) \frac{\sin 2\theta}{1 - \cos 2\theta} = \frac{2 \sin \theta \cos \theta}{1 - (1 - 2 \sin^2 \theta)}$$

$$= \frac{2 \sin \theta \cos \theta}{2 \sin^2 \theta}$$

$$= \frac{\cos \theta}{\sin \theta}$$

$$= \cot \theta$$

$$ii) \cot 15^\circ = \frac{\sin(2 \times 15^\circ)}{1 - \cos(2 \times 15^\circ)}$$

$$= \frac{\sin 30^\circ}{1 - \cos 30^\circ}$$

$$= \frac{\frac{1}{2}}{1 - \frac{\sqrt{3}}{2}}$$

$$= \frac{1}{2 - \sqrt{3}}$$

$$c) \int 2x^2 e^{x^3} dx = \frac{2}{3} \int 3x^2 e^{x^3} dx$$

$$= \frac{2}{3} e^{x^3} + C$$

$$d) \text{P.T.O}$$

$$d) \quad \frac{d}{dx} \log_e \left(\frac{1+\sin x}{\cos x} \right)$$

(2)

$$= \frac{d}{dx} \left(\log_e (1+\sin x) - (\log_e \cos x) \right)$$

$$= \frac{\cos x}{1+\sin x} + \frac{+\sin x}{\cos x}$$

$$= \frac{\cos^2 x + \sin x + \sin^2 x}{\cos x (1+\sin x)}$$

$$= \frac{(1+\sin x)}{\cos x (1+\sin x)}$$

$$= \sec x$$

$$\begin{aligned} e) \quad \sqrt{1-2\sin x} &= \sqrt{\cos^2 x + \sin^2 x - 2\sin x \cos x} \\ &= \sqrt{\cos^2 x - 2\sin x \cos x + \sin^2 x} \\ &= \sqrt{(\cos x - \sin x)^2} \\ &= \pm (\cos x - \sin x) \end{aligned}$$

(3)

Now for $0 < x < \frac{\pi}{4}$, $\cos x > \sin x$

$$\begin{aligned} \therefore \int (\cos x - \sin x) dx \\ = \sin x + \cos x + C \end{aligned}$$